

Faia–Lochner–Schoefer (2026): from primitives to the treatment intensity D_i

Self-contained, line-by-line derivation of the composite markdown,
Eq. (21), and D_i

Markdown convention (read this first)

Faia define the markdown as $\xi \cdot MRPL = w$, i.e. (we write $MRPL$ throughout;
Faia denote it $MRPL$, using N for the labour input — same object)

$$\xi = \frac{w}{MRPL} \leq 1, \quad \xi = 1 \text{ competitive, } \xi < 1 \text{ monopsony (wage below marginal product).}$$

This is the **reciprocal** of the standard / proposal convention $\mu = MRPL/w \geq 1$,
so $\xi = 1/\mu$ and $\ln \xi = -\ln \mu$. A binding minimum wage raises w toward $MRPL$,
compressing the markdown up to $\xi = 1$ once the floor reaches $MRPL$.

i Note

Memo — diagram convention. The algebra below keeps Faia’s $\xi = w/MRPL \leq 1$. The *figures* instead label the markdown as $\mu^X = MRPL/w = 1/\xi$ (so $\mu^X = 1$ competitive, $\mu^X > 1$ markdown), matching the proposal’s μ^X for the SARS application. Read $\mu^X = 1 \Leftrightarrow \xi = 1$ and $\mu^X > 1 \Leftrightarrow \xi < 1$.

i Note

Faia **measure** the markdown with the production-approach ratio (De Loecker/Morlacco), assuming no wedge in intermediate inputs ($\mu^M = 1$) — the same assumption as the proposal. Only the **test** (does a binding minimum wage bring ξ to 1?) is design-based.

Why $\xi = w/MRPL$ and not the conventional $\mu = MRPL/w$?

Both are reciprocals of the same wedge; the merit is in what each makes visible. With firm labour-supply elasticity ε (Faia's θ , from $n^s = w^\varepsilon$):

$$\xi = \frac{w}{MRPL} = \frac{\varepsilon}{1 + \varepsilon} \in (0, 1], \quad \mu = \frac{1}{\xi} = \frac{MRPL}{w} = \frac{1 + \varepsilon}{\varepsilon} \in [1, \infty).$$

object	Faia $\xi = w/MRPL$	conventional $\mu = MRPL/w$
structural form	$\xi = \frac{\varepsilon}{1 + \varepsilon}$ (pass-through: $w = \xi MRPL$)	$\mu = \frac{1 + \varepsilon}{\varepsilon}$
range	bounded $(0, 1]$	unbounded $[1, \infty)$
competitive	$\xi = 1$, $\ln \xi = 0$ — a ceiling	$\mu = 1$, $\ln \mu = 0$ — a floor
“monopsonistic” flag	$\ln \xi < 0$	$\ln \mu > 0$
type $\rightarrow$ firm (Eq. 19)	$\xi_i^{-1} = \sum_o \phi_o \xi_o^{-1}$ — linear in the inverse (harmonic in ξ)	$\mu_i = \sum_o \phi_o \mu_o$ — linear directly (arithmetic)
exposed $\rightarrow$ competitive	$\Delta \ln \xi = -\ln \xi \ (\rightarrow 0^-)$, the “gap from one” (Faia Eq. 17)	$\Delta \mu = 1 - \mu_E \ (\rightarrow 1^+)$
treatment intensity	$D_i = -\Phi_E \ln \xi \geq 0$	$D_i = \Phi_E \ln \mu \geq 0$
DiD benchmark	$\rho = 1$	$\rho = 1$

Reading the columns:

- **ξ 's edge (why Faia pick it):** bounded and *signed* — monopsony $\Leftrightarrow \ln \xi < 0$ (their “2013 log markdown below zero” subsample), the reform is an absorbing **ceiling** at $\ln \xi = 0$, treatment $D_i = -\Phi_E \ln \xi \geq 0$ is a non-negative “distance to competitive,” and $\xi = \varepsilon/(1 + \varepsilon)$ is the structural pass-through itself.
- **μ 's edge:** it aggregates linearly *without* a reciprocal — Eq. (19) is a bare payroll-weighted arithmetic mean.
- They are reciprocals, so Faia **read** results in ξ and **compute** in $\xi^{-1} = \mu$ — no trade-off. Hence “linear only in the inverse” is a statement *about* ξ , not a reason to prefer ξ over μ .

Notation (all firm i , worker types / occupations o)

symbol	meaning
$n_{o,i}$	employment of type o
$w_{o,i}$	wage of type o
$MRPL_{o,i}$	marginal revenue product of type o
$\xi_{o,i} = w_{o,i}/MRPL_{o,i} \leq 1$	type-level markdown; inverse
	$\xi_{o,i}^{-1} = MRPL_{o,i}/w_{o,i} \geq 1$
$\kappa_{o,i} = n_{o,i}/\sum_{o'} n_{o',i}$	employment weight
$\phi_{o,i} = \frac{w_{o,i} n_{o,i}}{\sum_{o'} w_{o',i} n_{o',i}}$	wage-bill (payroll) weight

Type-level wage-setting condition (just a rearrangement of the markdown definition):

$$MRPL_{o,i} = w_{o,i} \xi_{o,i}^{-1}. \quad (*)$$

The composite (firm-level) markdown

Define a firm mean wage and a composite marginal product, both employment-weighted:

$$w_i = \sum_o \kappa_{o,i} w_{o,i}, \quad MRPL_i = \sum_o \kappa_{o,i} MRPL_{o,i}.$$

Define the composite markdown ξ_i so the firm mimics a single worker (Faia Eq. 18):

$$w_i \xi_i^{-1} = MRPL_i \quad \Longleftrightarrow \quad \xi_i^{-1} = \frac{MRPL_i}{w_i}. \quad (18)$$

Firm-level aggregation — Eq. (19): the authors use ξ^{-1} , but the conventional μ^X aggregates identically

Start from (18) and substitute (*) into $MRPL_i$:

$$\xi_i^{-1} = \frac{MRPL_i}{w_i} = \frac{\sum_o \kappa_{o,i} MRPL_{o,i}}{w_i} = \frac{\sum_o \kappa_{o,i} w_{o,i} \xi_{o,i}^{-1}}{w_i}.$$

Now simplify the weight $\frac{\kappa_{o,i} w_{o,i}}{w_i}$. Using $w_i = \frac{\sum_o n_{o,i} w_{o,i}}{\sum_o n_{o,i}}$ and $\kappa_{o,i} = \frac{n_{o,i}}{\sum_o n_{o,i}}$:

$$\frac{\kappa_{o,i} w_{o,i}}{w_i} = \frac{\frac{n_{o,i}}{\sum n} w_{o,i}}{\frac{\sum_{o'} n_{o',i} w_{o',i}}{\sum n}} = \frac{n_{o,i} w_{o,i}}{\sum_{o'} n_{o',i} w_{o',i}} = \phi_{o,i}.$$

The $\sum n$ cancels, and what is left is exactly the payroll weight. Therefore

$$\boxed{\xi_i^{-1} = \sum_o \phi_{o,i} \xi_{o,i}^{-1}} \quad (19)$$

Equivalently, since $\mu^X = \xi^{-1} = MRPL/w$, the conventional markdown obeys the identical payroll-weighted linear aggregation:

$$\boxed{\mu_i^X = \sum_o \phi_{o,i} \mu_{o,i}^X} \quad (19')$$

In words: the firm's *inverse* markdown — equivalently the conventional μ^X — is the payroll-weighted average of the workers' values. The authors use the inverse, but the aggregation is linear in either representation; only ξ_o itself (not its reciprocal μ^X) fails to average linearly. “Inverse form” is a representation choice, not a necessity.

Split workers into exposed and non-exposed — Eq. (20)

Partition types into exposed E (pre-reform wage below the coming floor) and non-exposed N . Just break the single sum in (19) into two:

$$\xi_i^{-1} = \underbrace{\sum_{o \in E} \phi_{o,i} \xi_{o,i}^{-1}}_{\Phi_{E,i} \xi_{E,i}^{-1}} + \underbrace{\sum_{o \in N} \phi_{o,i} \xi_{o,i}^{-1}}_{\Phi_{N,i} \xi_{N,i}^{-1}} = \Phi_{E,i} \xi_{E,i}^{-1} + \Phi_{N,i} \xi_{N,i}^{-1}, \quad (20)$$

where the **payroll share of exposed workers** and the **within-group average inverse markdown** are

$$\Phi_{E,i} = \sum_{o \in E} \phi_{o,i}, \quad \xi_{E,i}^{-1} = \sum_{o \in E} \frac{\phi_{o,i}}{\Phi_{E,i}} \xi_{o,i}^{-1} \quad (\text{and analogously for } N).$$

Note $\Phi_{E,i} + \Phi_{N,i} = 1$.

Post-reform markdown — Eq. (21)

Model restriction. $\mu^X = 1$ does **not** hold in general — only for a subset of exposed firms (see Figure 3). For that subset, a binding minimum wage lifts the exposed type’s wage up to its marginal product, so its markdown becomes competitive:

$$\xi'_o = 1 \quad \forall o \in E \quad \implies \quad \xi'^{-1}_E = \sum_{o \in E} \frac{\phi_{o,i}}{\Phi_E} \cdot 1 = 1.$$

Non-exposed unchanged, $\xi'^{-1}_N = \xi^{-1}_N$, and payroll shares Φ_E, Φ_N are assumed stable. Substitute into (20):

$$\boxed{\xi'^{-1}_i = \Phi_E \cdot 1 + \Phi_N \xi'^{-1}_N} \tag{21}$$

Read it directly: after the reform, the exposed slice of the payroll (Φ_E) carries an inverse markdown of exactly 1 (competitive), and the remaining slice (Φ_N) keeps its old inverse markdown. Eq. (21) is written on ξ^{-1} only by the authors’ choice; since $\xi^{-1} = \mu^X = MRPL/w$ adds linearly, the identical equation holds directly in μ^X : $\mu^{X'}_i = \Phi_E \cdot 1 + \Phi_N \mu^X_N$. The “inverse form” carries no special content.

When is $\xi'_E = 1$ exact? The binding-floor geometry

Eq. (21) sets every exposed type to $\xi' = 1$ (i.e. $\mu^X = 1$, $w = MRPL$) only when the floor $\bar{w}$ is high enough relative to the monopsony and competitive wages w_M, w_c :

$$\begin{aligned} \text{(a)} \quad w_M < \bar{w} < w_c &\implies \mu^X > 1 \\ \text{(b)} \quad \bar{w} = w_c &\implies \mu^X = 1 \\ \text{(c),(d)} \quad \bar{w} > w_c &\implies \mu^X = 1 \\ \text{(e)} \quad \bar{w} < w_M < w_c &\implies \mu^X > 1 \end{aligned}$$

Common points across the panels: **A** monopsony ($w = w_M$); **B** the firm’s operating (post-floor) employment; **C** competitive; **E** the marginal product at monopsony employment L_M , i.e. $w_E = MRPL(L_M)$. No colour fills — the exposed range $[0, L_M]$ is a bracket on the L -axis (and the “lost employment” range in (c)/(d)). ($MRPL = MRPN$ in Faia.)

The five regimes, ordered by the floor $\bar{w}$, are drawn in Figure 1.

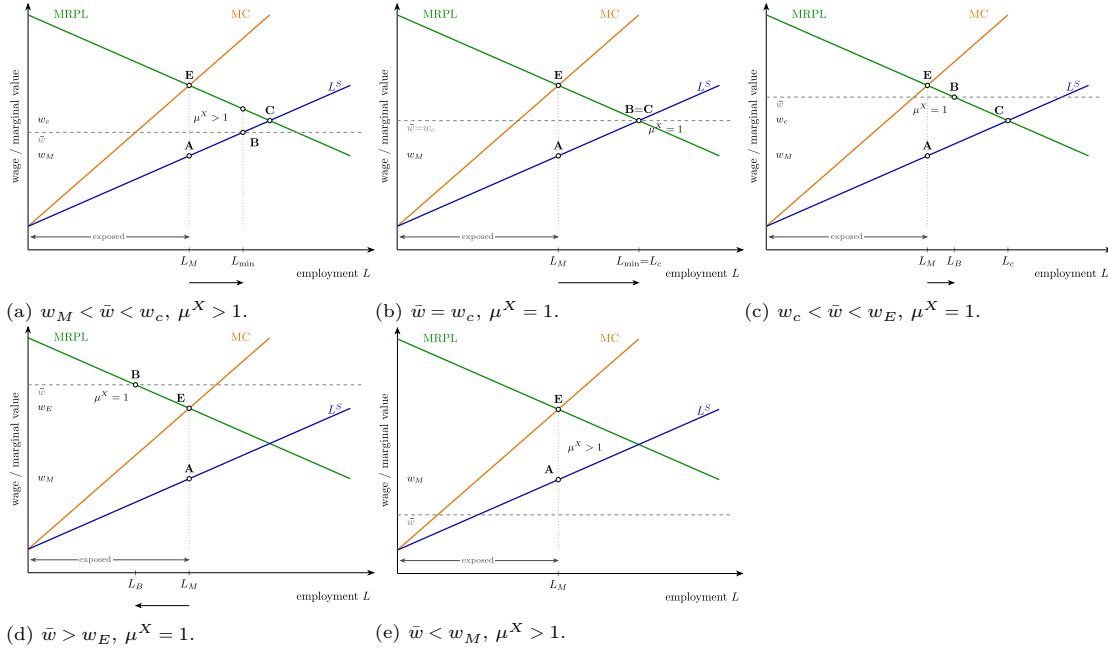


Figure 1: Five regimes of the floor $\bar{w}$. Eq. (21)'s $\mu_E^X = 1$ holds in (b), (c), (d); (a) and (e) $\mu^X > 1$, so Eq. (21) overstates compression, a possible reason for $\rho \leq 0.25$.

Identifying the clean subset: the employment-growth cut (Panel (b))

$\mu^X = 1$ does not hold in general — only for a subset of exposed firms; see Figure 3a, Figure 3b, Figure 3c.

Faia target the exposed firms that

- shrink = panel (a)
- of below-median employment growth = panel (b) ($\mu^X = 1$) and (c) ($\mu^X > 1$)

Quotes backing the logic:

(p. 7) “we will tag one unambiguous subset of exposed firms ... exposed firms that shrink or have below-median employment growth—those to the left of point B in Panel (b).”

(p. 28) “we split firms by their employment growth into the reform (log employment in 2016 minus that in 2013).”

(p. 30) “inherently tentative, as it assumes that the employment growth is not caused by negative shifts in the labor supply curve.”

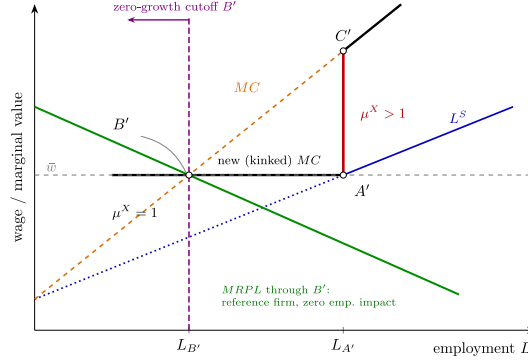


Figure 2: Reference firm with zero employment impact.

The four exposed-firm outcomes, ordered left-to-right by the firm's *MRPL* position. E' is where monopsony employment is determined; F' is employment under the minimum wage:

From Eq. (21) to the markdown change — Eq. (24)

Subtract pre-reform (20) from post-reform (21). The non-exposed term is identical in both, so it cancels:

$$\xi_i'^{-1} - \xi_i^{-1} = \Phi_E(1 - \xi_E^{-1}).$$

Take the main-text case where the exposed group is representative of the firm, $\xi_E = \xi_i$:

$$\Delta(\xi_i^{-1}) = \Phi_E(1 - \xi_i^{-1}). \quad (\dagger)$$

Convert the change in the inverse into the proportional change in the level. Two facts:

$$\Delta(\xi^{-1}) = \Phi_E(1 - \xi^{-1}) = \Phi_E \frac{\xi - 1}{\xi} \quad (\text{from } (\dagger)), \quad \Delta(\xi^{-1}) \approx -\frac{\Delta\xi}{\xi^2} \quad (\text{first order}).$$

Equate the two:

$$-\frac{\Delta\xi}{\xi^2} = \Phi_E \frac{\xi - 1}{\xi} \implies \Delta\xi = -\xi \Phi_E(\xi - 1) = \xi \Phi_E(1 - \xi).$$

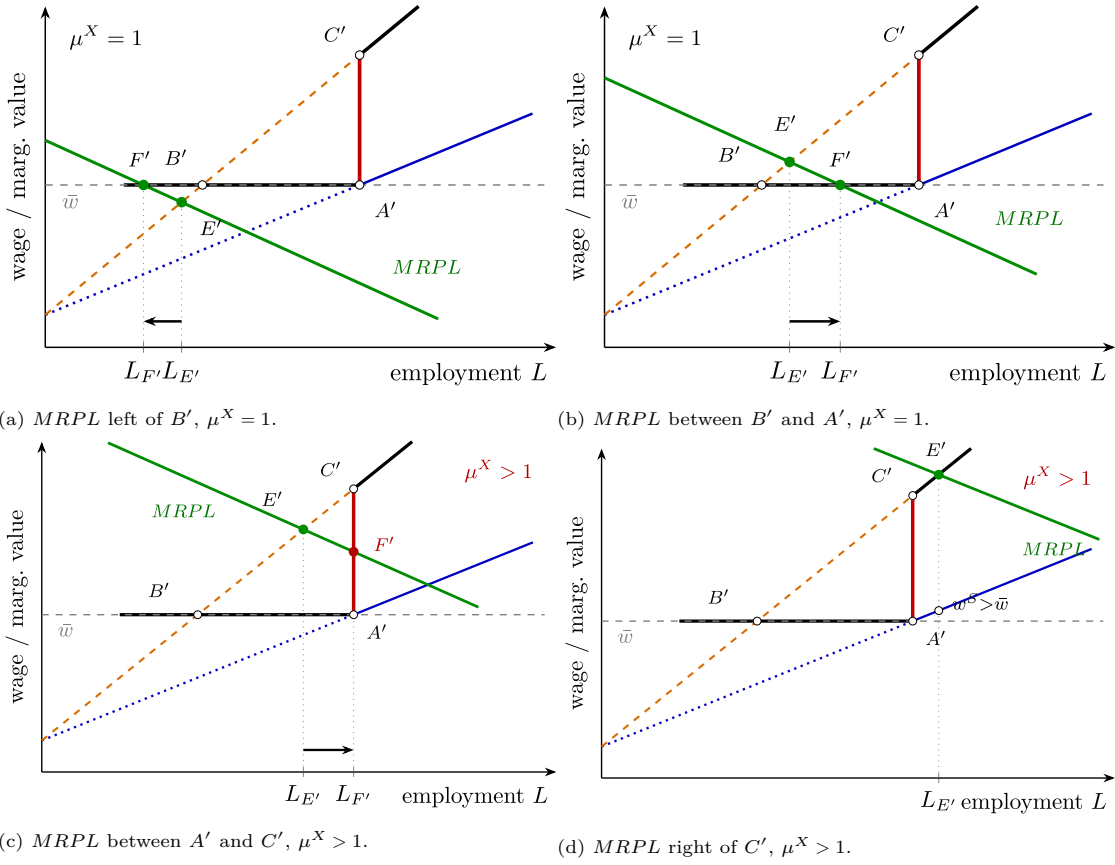


Figure 3: $\mu^X = 1$ holds only in (a) (shrinks) and (b) (grows, but employment stays below $L^S(\bar{w})$).

Divide by ξ :

$$\boxed{\frac{\Delta \xi}{\xi} = \Phi_E (1 - \xi)} \quad (24)$$

(The general case keeping $\xi_E \neq \xi$ is Faia's footnote 9.)

Including $\mu^X > 1$ firms

Include a subset with $\mu^X > 1$: $\xi_E'^{-1} > 1$, so the exposed slice no longer maps to 1 and (24) becomes

$$\frac{\Delta \xi}{\xi} = \Phi_E \left(1 - \frac{\xi}{\xi'} \right) = \underbrace{-\Phi_E \ln \xi}_{D_i} + \underbrace{\left(-\Phi_E \xi (1/\xi' - 1) \right)}_{w_i}.$$

In $\Delta \ln \xi_i = \rho D_i + v_i$ with w_i omitted, $\hat{\rho} = 1 + \text{Cov}(w_i, D_i) / \text{Var}(D_i)$. We see $\text{Cov}(w_i, D_i) < 0$, because

$$w_i = -\Phi_E \xi (1/\xi' - 1) \leq 0, \quad \frac{\partial |w_i|}{\partial (-\ln \xi)} > 0, \quad D_i = -\Phi_E \ln \xi,$$

so $|w_i|$ and D_i co-move: $\text{Cov}(w_i, D_i) = -\text{Cov}(|w_i|, D_i) < 0$, giving $\hat{\rho} < 1$ ($\rho \leq 0.25$). The clean subset sets $w_i = 0$, restoring $\rho = 1$.

From Eq. (24) to D_i — Eqs. (25)–(27)

Log-linearize using $\frac{\Delta \xi}{\xi} \approx \Delta \ln \xi$ and $\ln \xi \approx \xi - 1$ (exact as $\xi \rightarrow 1$):

$$1 - \xi \approx -\ln \xi \implies \Delta \ln \xi \approx -\Phi_E \ln \xi. \quad (25)$$

Write this as a difference-in-differences regression across firms and read off the treatment intensity:

$$\Delta \ln \xi_i = \rho D_i + v_i, \quad \boxed{D_i = -\Phi_{i,E} \ln \xi_i}. \quad (26-27)$$

D_i = (payroll share exposed) $\times$ (pre-reform markdown gap $-\ln \xi$), from pre-reform (2013) values.

Benchmark prediction: $\rho = 1$. **Empirically** $\rho \in [0, 0.25]$ — markdowns move only 0–25% of the predicted amount.

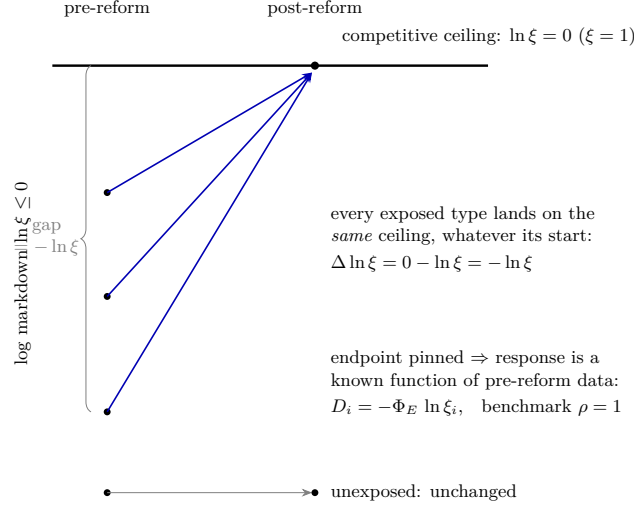


Figure 4: **Why the DiD is clean.** The markdown lives on $\ln \xi \leq 0$ with an absorbing **ceiling** at $\ln \xi = 0$ (competitive, $\xi = 1$). A binding floor lifts every *exposed* type to that same ceiling whatever its pre-reform depth, so the predicted change is the entire gap, $\Delta \ln \xi = -\ln \xi$; weighting by exposed payroll share gives $D_i = -\Phi_E \ln \xi_i$ with benchmark $\rho = 1$. Unexposed types are unchanged.

i Note

The exact relation is nonlinear in D with $0 \leq D < 1$; since $D \ll 1$ in the data, the linear approximation is tight and, if anything, slightly biased **toward** the monopsony prediction.

Why correlate Φ_E with $-\ln \xi$, not with ξ ?

$-\ln \xi$ is the **actual factor** of the treatment, $D_i = \Phi_E \cdot (-\ln \xi)$, so $\text{Cov}(\Phi_E, -\ln \xi)$ is the covariance that governs $\text{Var}(D_i)$. Correlating with ξ would diagnose a decomposition they never use.

Near $\xi \approx 1$: $-\ln \xi \approx 1 - \xi$, so $\text{corr}(\Phi_E, -\ln \xi) \approx -\text{corr}(\Phi_E, \xi)$ — same magnitude, opposite sign (as $-\ln \xi$ is decreasing in ξ). The log changes magnitude only for deep markdowns (ξ far below 1). German markdowns sit near 1, so the transform is near-innocuous.

What they report: $\text{corr}(\Phi_E, -\ln \xi) \approx 0.020$ (binned slope ≈ 0.076); variance decomposition attributes $\approx 83\%$ (monopsonistic sample) / $\approx 48\%$ (full) of $\text{Var}(D_i)$ to Φ_E , with negligible covariance.

Implication for the SARS application

Our μ^X ranges **well above 1** (unlike Germany), so level vs. log will materially differ. Report **both** $\text{corr}(\text{exposure}, \mu^X)$ and $\text{corr}(\text{exposure}, \ln \mu^X)$, plus the variance decomposition. The channel to worry about: $\mu^X \propto \theta_X / \alpha^X$, so labour-intensive (high α^X) firms are more exposed **and** carry lower measured markdown — a confound Germany did not face. Do not let a single transform decide whether it is “small.”